Adaptive CLIK Control for Robotic Arm Capture of Tumbling Satellites Via Genetic Fuzzy Trees

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Abstract. As humanity's presence in Earth's orbit grows, so too does the amount of space debris. To ensure that there is proper room in Earth's orbits to continue scientific missions and to allow for interplanetary travel, there needs to be a widespread method to manage debris autonomously. This work explores the use of Fuzzy Logic combined with the widely-used CLIK control to perform tumbling satellite capture safely and efficiently. By dynamically tuning the CLIK's K_P, K_D, p_i, and w_{jk} gains using Genetic Fuzzy Trees, performance can be optimized without compromising safety. It was found that the Genetic Fuzzy System performs very well in its training case reducing both settling time and control effort compared to standard CLIK control without violating joint constraints or colliding, but it requires further training on more cases to be generalized better.

Keywords: Space Robotics · AI · Fuzzy Logic.

1 Introduction

The increasing utilization of satellites in Low Earth Orbit (LEO) for GPS, communication, and climate analysis, among other applications, has led to a rise in orbital debris resulting from failed missions and satellites unable to deorbit. This growth highlights the necessity for autonomous systems capable of managing debris effectively [1]. Robotic manipulators attached to satellites, collectively termed Space Manipulator Systems (SMS), have emerged as viable solutions for capturing and managing non-cooperative satellites. Successful SMS operation involves three phases: rendezvous, capture, and stabilization [2,3].

This paper specifically investigates the capture phase, similar to the study by Karthikeyan and Kim [2], but differs by utilizing a two-dimensional system with a Fuzzy Logic-enhanced closed-loop inverse kinematics (CLIK) control approach [4,5]. The incorporation of Fuzzy Logic allows tuning of additional gains, aiming to improve controller performance and maintain safety. Moreover, the fuzzy logic-based approach introduces greater explainability, potentially enhancing user trust in autonomous systems. This paper details the control design methodology and compares its performance against a standard CLIK controller.

2 System Dynamics

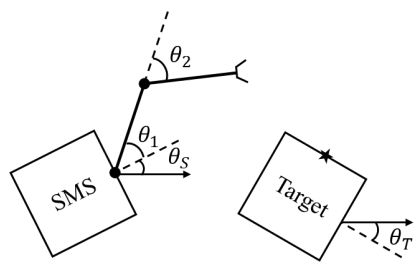

Fig. 1. Visualization of Tumbling Satellite Capture with Robotic Manipulator.

The system being examined can be seen in Figure 1. Here, the generalized coordinates considered are the angles at which SMS, the robotic manipulator links, and the target can rotate. These are encapsulated in the θ_S, θ_1, θ_2, and θ_T terms respectively.

To derive the dynamic equations, the positions of each body must be considered. As was previously stated, it is assumed that rendezvous has already occurred, so the positions of the SMS, $\mathbf{r_S}$, and target satellite, $\mathbf{r_T}$, are predefined and constant throughout capture. The points being considered on the robotic manipulators are the link midpoints which are calculated in (1).

$$\mathbf{r}_{B_i} = \mathbf{r}_{B_{i-1}} + \mathbf{C}_{NS}\mathbf{C}_{SB_{i-1}}\left([\rho_{i-1},\ 0]^T + \mathbf{C}_{B_{i-1}B_i}[\rho_i,\ 0]^T\right),\ i = 1, 2 \tag{1}$$

Here, $\mathbf{C}$ is the directional cosine matrix, N refers to the inertial frame, S refers to the SMS frame, and B_i refers to the ith body frame. Additionally, ρ_i is half the length of the ith body. It should be noted that $i = 0$ refers to the SMS and $i = 3$ refers to the target satellite.

The velocity of the links can then be calculated by taking the time derivative of the position expressions in (1). With the positions and velocities of the bodies defined, the Lagrangian for the dynamic system can be formed using the equation $L = T - V$, where L is the Lagrangian, T is the kinetic energy of the system, and V is the potential energy of the system. Notably, $V = 0$ for this case since gravitational effects are considered negligible in this space environment.

T contains translational and rotational components, meaning $T = T_{trans} + T_{rot}$, where $T_{trans} = \frac{1}{2}\sum_{i=0}^{3} m_i \mathbf{v}_i^T \mathbf{v}_i$ and $T_{rot} = \frac{1}{2}\sum_{i=0}^{3} I_i \dot{\theta}_i$. The dynamic equations can then be derived by solving the Euler-Lagrange equation, $\frac{\partial L}{\partial \mathbf{q}} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\mathbf{q}}}\right) = 0$, where $\mathbf{q} = [\theta_S\ \theta_1\ \theta_2\ \theta_T]$.

3 Control Methodology

3.1 CLIK Control

CLIK is a control technique most widely used for kinematically redundant systems [5]. A kinematically redundant system is one that has more degrees of freedom than what is required to complete a task. Here, in the two-dimensional system formulated before, the SMS has three degrees of freedom (satellite angle, first link angle, and second link angle), and the capture point has two degrees of freedom (x and y position). Hence, the only information that is required to formulate this control law is the end-effector Jacobian with respect to the SMS states.

While CLIK is not the only control technique that can be used for this application, its use in [2] inspired the authors to attempt to form a more robust framework using Genetic Fuzzy Trees. For the sake of space, a full derivation of CLIK control is omitted. Instead, one should refer to Chiacchio et al. [5].

CLIK control focuses on two objectives, a primary objective and a secondary objective. This provides an overall desired acceleration, and then inverse kinematics are used to produce the appropriate torques. The desired acceleration can thus be modeled using $\ddot{\mathbf{q}} = \ddot{\mathbf{q}}_P + \ddot{\mathbf{q}}_S$ where $\ddot{\mathbf{q}}_P$ is defined in (2) and $\ddot{\mathbf{q}}_S$ is defined in (3).

$$\ddot{\mathbf{q}}_P = J_e^+ \left(\ddot{\mathbf{x}}_{des} + K_P \mathbf{e} + K_D \dot{\mathbf{e}} - \dot{J}_e \dot{\mathbf{q}} \right) \tag{2}$$

$$\ddot{\mathbf{q}}_S = \left(I - J_e^+ J_e \right) \left(\dot{\boldsymbol{\lambda}} + K_N \dot{\mathbf{e}}_N \right) - \left(J_e^+ \dot{J}_e J_e^+ + \dot{J}_e^+ \right) J_e \tag{3}$$

In (2) and (3), J_e is the Jacobian of the end-effector and J_e^+ is the psuedo-inverse of J_e. (2) is a proportional derivative control law for the end-effector relative to the capture point, so K_P is the proportional gain and K_D is the derivative gain. Additionally, $\mathbf{e} = \mathbf{x}_{des} - \mathbf{x}_e$ where $\mathbf{x}_e$ is the end-effector cartesian position vector.

In (3), $\dot{\mathbf{e}}_N = \left(I - J_e^+ J_e \right) \left(\boldsymbol{\lambda} - \dot{\mathbf{q}} \right)$ is the null-space velocity error, and K_N is a positive gain that aids the tracking error to zero in the null space. Furthermore, $\lambda = -\frac{\partial \gamma}{\partial \mathbf{q}}$ where γ is a scalar function that defines the secondary objective. Here, $\gamma = \gamma_1 + \gamma_2$, which are defined in (4) and (5). Without going into too much detail, γ_1 enforces physical angle limit constraints for joints, and γ_2 enforces collision constraints for the robotic arm and the target satellite. p_i and w_{jk} are positive weights that dictate how much a constraint is enforced, and three critical points are considered with the two nearest obstacle points for the γ_2 calculation. Those critical points are the end-effector, the center of Link 2, and the start of Link 2.

$$\gamma_1 = \frac{1}{3} \sum_{i=1}^{3} p_i \left(\frac{q_i - \bar{q}_i}{\max(q_i) - \min(q_i)} \right)^2 \tag{4}$$

$$\gamma_2 = \sum_{j=1}^{3} \sum_{k=1}^{2} w_{jk} \frac{1}{d_{jk}^2} \tag{5}$$

3.2 Fuzzy-CLIK Control

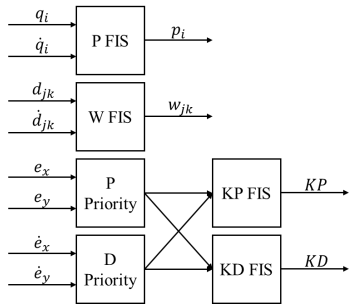

Fig. 2. Fuzzy Inference Systems to Tune CLIK Gains.

Figure 2 shows the Genetic Fuzzy Tree (GFT) designed to dynamically tune the weights p_i, w_{jk}, K_P, and K_D. Each fuzzy inference system (FIS) seen in this diagram utilizes Takagi-Sugeno defuzzification [7]. Both the P FIS and the W FIS are shared among each critical point/obstacle point pair for ease of computation and optimization.

Each FIS in Figure 2 was defined with three membership functions. These membership functions are evenly distributed throughout the input ranges and are kept constant throughout the training process so that the polynomial rule coefficients can be trained to produce the desired behavior. P FIS, P Priority, and D Priority all took normalized inputs on the range $[-1, 1]$, whereas W FIS took inputs on the range $[0, 1]$ since d_{jk} and $\dot{d}_{jk}$ are simply relative distance and relative velocity magnitudes. The output of each FIS is originally on the range $(0, 1]$ before scaling is applied. Along with the rules, the output bounds for each FIS are added to the chromosome to be trained.

The specific implementation of this GFT utilized Thales' TrUE AI toolkit ™[6], which specializes in the design and training of GFTs. TrUE AI ™ has its own Genetic Algorithm (GA) that optimizes the defined GFT. This is a modified GA that is capable of tuning hyperparamters such as mutation rate, elitism rate, etc., so all that must be specified are the population size, the number of cores, and the cost function, shown in (6). In this work, 256 chromosomes are used for training over 16 cores.

$$cost = 5000 * T_s + ISC + \Delta \tag{6}$$

where $ISC \equiv \sum_{i=1}^{N} \boldsymbol{\tau}(t_i)\boldsymbol{\tau}(t_i) * \Delta t$ is the integral over the squared control. In (6), Δ is representative of a large penalty term for when undesirable behavior occurs (such as collisions, numerical instability, etc.), and 5000 was chosen as the weight for settling time, T_s, because it scaled that value to be in a similar order of magnitude as ISC making the multi-objective optimization more fair for both terms.

4 Results & Discussion

The system parameters used were $\rho_i = 0.5$ m, $m_0 = 250$ kg, $m_1 = m_2 = 25$ kg, $m_3 = 180$ kg, $I_0 = 25$ kg m^2, $I_1 = I_2 = 2.5$ kg m^2, and $I_3 = 18$ kg m^2[2].

Table 1. Training and Test Case Parameters.

Description	Training Case	Test Case
Central SMS Position (r_s)	$[-1.25, 0]$ m	$[-1.25, 0]$ m
SMS Initial Angles ($\mathbf{q}(0)$)	$[\frac{\pi}{3}, -0.3, -0.1]$ rad	$[0, -\frac{\pi}{3}, -\frac{\pi}{2}]$ rad
SMS Initial Angular Velocities ($\dot{\mathbf{q}}(0)$)	$[0, 0, 0]$ rad/s	$[0, 0, 0]$ rad/s
Central Target Position (r_t)	$[1.25, 0]$ m	$[1.25, 0]$ m
Target Initial Angle ($\theta_T(0)$)	0 rad	$\frac{\pi}{2}$ rad
Target Initial Angular Velocity ($\dot{\theta}_T(0)$)	0.2 rad/s	-0.3 rad/s

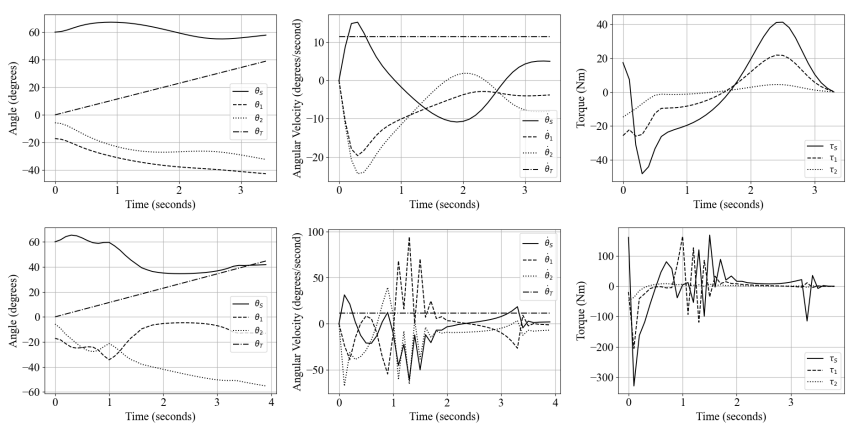

Fig. 3. Train Case State and Control History (Top is Trained GFT, Bottom is Standard CLIK).

Table 1 shows the parameters that were used in both the training and testing cases. These parameters were inspired from those used in [2] for their ability to showcase typical capture states. Gains were found for standard CLIK control by trial-and-error rather than through a search algorithm. It was found that in the training case shown in Figure 3, the settling time for standard CLIK control was 3.9 seconds with an ISC of 404,395 N^{2}m^2, while Fuzzy-CLIK achieved a settling time of 3.4 seconds with an ISC of just 28,137 N^{2}m^2. However, in the test case shown in Figure 4, standard CLIK's settling time was 3.2 seconds with an ISC of 7,021,986 N^{2}m^2, while Fuzzy-CLIK had a settling time of 4.1 seconds with an ISC of 748,351 N^{2}m^2.

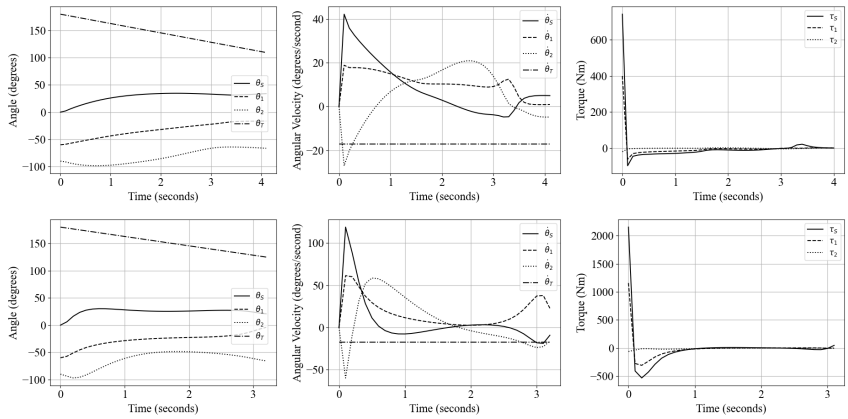

Fig. 4. Test Case State and Control History (Top Row is Trained GFT, Bottom is Standard CLIK).

5 Conclusion

This work showed how GFTs can be combined with the CLIK control scheme to achieve capture of a tumbling satellite in a 2D environment efficiently and quickly compared to standard CLIK control. For future work, standard CLIK gains should be obtained with more sophisticated methods for a more robust comparison, more cases should be used for training and testing, and higher fidelity simulations should be used for better comparisons.

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